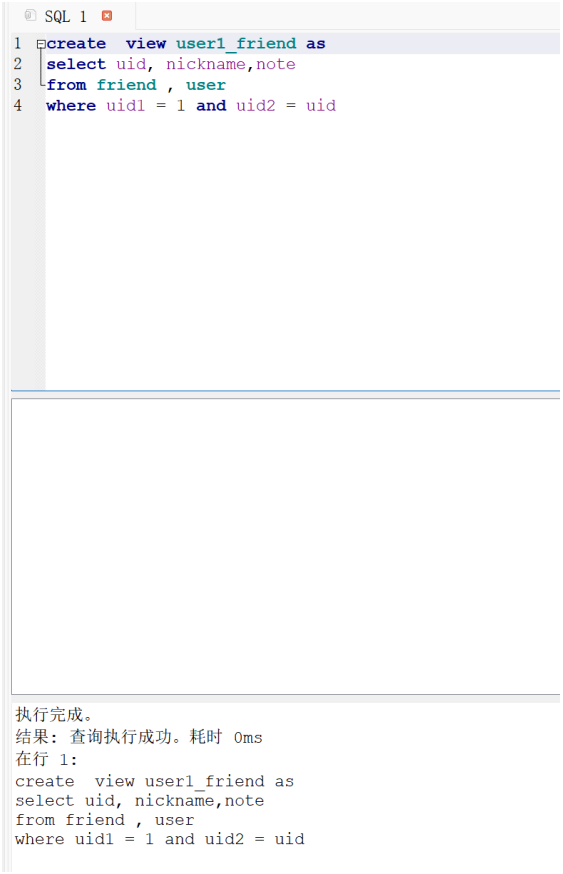
1

```
create view user1_friend as
select uid, nickname,note
from friend , user
where uid1 = 1 and uid2 = uid
```



2

```
select nickname, content,post_time
from user1_friend, moment using(uid)
where type = '公开' or type = '仅好友可见'
order by post_time desc
```

```sql
select nickname, content,post_time
from user1_friend, moment using(uid)
where type = '公开' or type = '仅好友可见'
order by post_time desc
```

	nickname	content	post_time
1	ahjlreh	csjmgsrvtqventguoo···	999
2	nndupcc	hnnzhjocmxkxomlpix···	998
3	dplshmv	rbkagugiervuapjqwc···	996
4	uxziiv	iulrumdtgbhrlazpfq···	996
5	gbuntfp	xacpxbfqepdtigvbrg···	994
6	qbydpagxl	dyyenccwpmifmvynvw···	993
7	fkhdduk	dpkkilbsuqsyaamsbv···	993
8	akdrqbcec	dzuvauthhgurakqske···	992
9	akdrqbcec	yyuzimppehtmoztknv···	989

3

select gname, content, sent_time
from group_send join groups using(gid) join message using(mid)
where uid = 1
order by gname, sent_time desc

```sql
select gname, content, sent_time
from group_send join groups using(gid) join message using(mid)
where uid = 1
order by gname, sent_time desc
```

	gname	content	sent_time
1	htbdloshcqogi	lipeobyzhaqqzspxbc	998
2	htbdloshcqogi	igakhmamtp	995
3	htbdloshcqogi	bytfnozgkdapqlho	994
4	htbdloshcqogi	ewmobiuahkxtuyphoik	992
5	htbdloshcqogi	jilxxfopki	992
6	htbdloshcqogi	senztvyiwqb	991
7	htbdloshcqogi	vf	991
8	htbdloshcqogi	hpouyrf	990
9	htbdloshcqogi	a	988

```
执行完成。
结果: 165 行返回，耗时 41ms
在行 1:
select gname, content, sent_time
from group_send join groups using(gid) join message using(mid)
where uid = 1
```

4.

```
select nickname, content, sent_time
from(
select sent_time, mid, uid_sender
from send
where send.uid_receiver = 1 and status = '待发送')
join message using(mid), user
where user.uid = uid_sender
order by uid_sender, sent_time DESC
```

SQL 1

```sql
select nickname, content, sent_time
from(
select sent_time, mid, uid_sender
from send
where send.uid_receiver = 1 and status = '待发送')
join message using(mid), user
where user.uid = uid_sender
order by uid_sender, sent_time DESC
```

	nickname	content	sent_time
1	sbycfmzog	vcvlfwnolmxuyecix	999
2	sbycfmzog	eyvlrwndvctv	999
3	sbycfmzog	t	997
4	sbycfmzog	vkb	986
5	sbycfmzog	bei	975
6	sbycfmzog	iictzttgubfnna	969
7	sbycfmzog	nawyfpdxavaluwi	968
8	sbycfmzog	tqnriccw	928
9	sbycfmzog	fqbi	912

执行完成。
结果: 8752 行返回，耗时 1753ms
在行 1:

5

```
select content, nickname from(
select mid, uid_sender
from send
where ((uid_receiver = 1 and uid_sender = 6)or(uid_receiver = 6 and uid_sender = 1))
and status = '已发送') m_u,
(select mid, content
from message
where content like '%晚安%') m_e, user
where user.uid = uid_sender and m_u.mid = m_e.mid
```

```sql
select content, nickname from(
select mid, uid_sender
from send
where ((uid_receiver = 1 and uid_sender = 6)or(uid_receiver = 6 and uid_sender = 1))
and status = '已发送') m_u,
(select mid, content
from message
where content like '%晚安%') m_e, user
where user.uid = uid_sender and m_u.mid = m_e.mid
```

	content	nickname
1	hghou晚安e	sonqoa
2	aygqqcoppocrfctfi晚…	sonqoa
3	omeshu晚安…	oilolc
4	dobsxjapfbj晚安…	oilolc
5	u晚安hwwtlcnkslyoxs	oilolc
6	bmcmbfs晚安wkgregf	oilolc

6.

create index send_uid_receiver on send(uid_receiver)

建立索引后，第四问的用时从 1730ms 减少到 54ms

建立索引后，第五问的用时从 718ms 减少到 4ms

理由：在第四问和第五问的都用到了 uid_receiver 的进行查询操作，且 send(uid_receiver)设计到的数据量大。所以对其建立索引。

SQL 1

```
1  create index send_uid_receiver on send(uid_receiver)
```

执行完成。
结果：查询执行成功。耗时 4536ms
在行 1:
create index send_uid_receiver on send(uid_receiver)

SQL 1

```sql
select nickname, content, sent_time
from(
select sent_time, mid, uid_sender
from send
where send.uid_receiver = 1 and status = '待发送')
join message using(mid), user
where user.uid = uid_sender
order by uid_sender, sent_time DESC
```

	nickname	content	sent_time
1	sbycfmzog	vcvlfwnolmxuyecix	999
2	sbycfmzog	eyvlrwndvctv	999
3	sbycfmzog	t	997
4	sbycfmzog	vkb	986
5	sbycfmzog	bei	975
6	sbycfmzog	iictzttgubfnna	969
7	sbycfmzog	nawyfpdxavaluwi	968
8	sbycfmzog	tqnriccw	928
9	sbycfmzog	fqbi	912

执行完成。
结果：8752 行返回，耗时 54ms

```sql
select content, nickname from(
select mid, uid_sender
from send
where ((uid_receiver = 1 and uid_sender = 6)or(uid_receiver = 6 and uid_sender = 1))
and status = '已发送') m_u,
(select mid, content
from message
where content like '%晚安%') m_e, user
where user.uid = uid_sender and m_u.mid = m_e.mid
```

	content	nickname
1	omeshu晚安…	oilolc
2	dobsxjapfbj晚安…	oilolc

```
执行完成。
结果: 6 行返回，耗时 4ms
在行 1:
select content, nickname from(
select mid, uid_sender
from send
where ((uid_receiver = 1 and uid_sender = 6)or(uid_receiver = 6 and uid_sender = 1))
and status = '已发送') m_u,
(select mid, content
from message
where content like '%晚安%') m_e, user
where user.uid = uid_sender and m_u.mid = m_e.mid
```

7.

```sql
select count(*)friend_num
from(
select uid
from joingroup
where gid = 1) g_u, friend
where uid1 = 1 and uid2 = g_u.uid
```

```
select count(*)friend_num
from(
select uid
from joingroup
where gid = 1) g_u, friend
where uid1 = 1 and uid2 = g_u.uid
```

	friend_num
1	11

8.

select user.nickname, ((user.longitude - user_one.longitude)*(user.longitude - user_one.longitude)+(user.latitude - user_one.latitude)*(user.latitude - user_one.latitude))
from user,
(select longitude, latitude
from user
where uid = 1
)user_one
where user.uid not in
(select uid2
from friend
where uid1= 1
)and ((user.longitude - user_one.longitude)*(user.longitude - user_one.longitude)+(user.latitude - user_one.latitude)*(user.latitude - user_one.latitude)) < 100



```
SQL 1
select user.nickname, ((user.longitude - user_one.longitude)*(user.longitude - user_one.longitude)+(user.latitude - user_one.latitude)*(user.latitude - user_one
from user,
(select longitude, latitude
from user
where uid = 1
)user_one
where user.uid not in
(select uid2
from friend
where uid1= 1
)and ((user.longitude - user_one.longitude)*(user.longitude - user_one.longitude)+(user.latitude - user_one.latitude)*(user.latitude - user_one.latitude)) < 100
```

	nickname	e.longitude)+(user.la
1	sonqoa	0
2	qfjavzte	72

9.

```
update message
set shared_count = shared_count - (
select count(*)
from send
where ((uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0))
and message.mid = send.mid
)
where mid in(
select mid
from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)
)
```

```sql
update message
set shared_count = shared_count - (
select count(*)
from send
where ((uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0))
and message.mid = send.mid
)
where mid in(
select mid
from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)
)
```

```
执行完成。
结果: 查询执行成功。耗时 238ms，120 行数据受影响
在行 1:
update message
set shared_count = shared_count - (
select count(*)
from send
where ((uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0))
and message.mid = send.mid
)
where mid in(
select mid
from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)
)
```

delete from friend
where ((uid1 = 1 and uid2 = 0)or(uid1 = 0 and uid2 = 1))

```sql
delete from friend
where ((uid1 = 1 and uid2 = 0)or(uid1 = 0 and uid2 = 1))
```

```
执行完成。
结果: 查询执行成功。耗时 0ms, 2 行数据受影响
在行 1:
delete from friend
where ((uid1 = 1 and uid2 = 0)or(uid1 = 0 and uid2 = 1))
```

delete from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)

```sql
delete from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)
```

执行完成。
结果：查询执行成功。耗时 764ms，127 行数据受影响
在行 1:

```sql
delete from send
where (uid_receiver = 0 and uid_sender = 1)
or (uid_receiver = 1 and uid_sender = 0)
```

delete from message
where shared_count = 0

```sql
delete from message
where shared_count = 0

```

执行完成。
结果：查询执行成功。耗时 22613ms，35 行数据受影响
在行 1：

```sql
delete from message
where shared_count = 0
```